

NITROGEN CONTAINING COMPOUND

1. IUPAC Name of $\text{CH}_3\text{CH}_2-\text{N}(\text{CH}_2\text{CH}_3)_2$ is—

- (1) N-Butyl-N-ethyl ethanamine
- (2) N, N-diethyl propanamine
- (3) N, N-di methyl butanamine
- (4) N, N-di ethyl butyl amine

2. $\text{C}_6\text{H}_5-\text{CH}_2-\text{Cl} \xrightarrow[\text{(excess)}]{\text{NH}_3} \text{'A'} \xrightarrow{2\text{CH}_3\text{Cl}} \text{'B'}$

Product 'B' is—

- (1) $\text{C}_6\text{H}_5-\text{CH}_2-\text{NH}-\text{CH}_3$
- (2) $\text{C}_6\text{H}_5-\text{CH}_2-\text{NH}_2$
- (3) $\text{C}_6\text{H}_5-\text{CH}_2-\text{N}(\text{CH}_3)_2$

- (4) $\text{C}_6\text{H}_5-\text{N}(\text{CH}_3)_2$

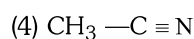
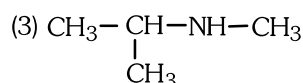
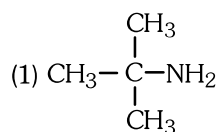
3. Which of the following compound gives propanamine on reaction with LiAlH_4 —

- (1) Ethyl cyanide
- (2) Propyl cyanide
- (3) Nitro ethane
- (4) Both (1) & (3)

4. Aniline can not be prepared by Gabriel phthalimide synthesis due to—

- (1) Bulky nature of Benzene Ring
- (2) Aryl halide do not undergo nucleophilic substitution by Phthalimide ion
- (3) Aryl halide do not undergo electrophilic substitution reaction
- (4) Both (1) & (3)

5. Select the Primary amine



6. Reactivity order of Alkyl halide towards ammonolysis is—

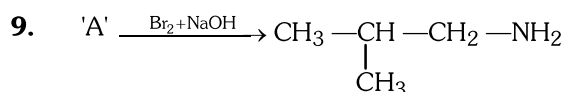
- (1) $\text{R}-\text{I} > \text{R}-\text{Br} > \text{R}-\text{Cl}$
- (2) $\text{R}-\text{Cl} > \text{R}-\text{Br} > \text{R}-\text{I}$
- (3) $\text{R}-\text{I} < \text{R}-\text{Br} > \text{R}-\text{Cl}$
- (4) $\text{R}-\text{I} > \text{R}-\text{Br} < \text{R}-\text{Cl}$

7. Which of the following can be used to distinguish ethyl amine from dimethyl amine —

- (1) Benzene sulphonyl chloride
- (2) Diethyl oxalate
- (3) p-Methyl benzene sulphonyl chloride
- (4) All of these

8. Which one can gives propanamine on reaction with LiAlH_4

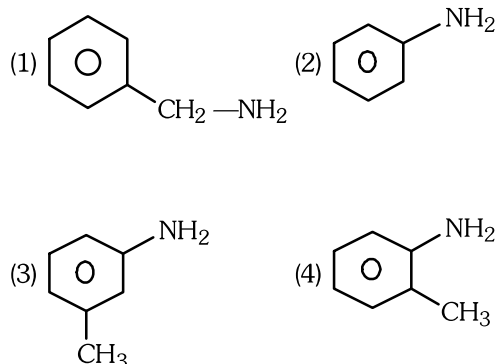
- (1) Propyl cyanide
- (2) N-Methyl Propanamide
- (3) Propane nitrile
- (4) Both (1) & (3)



Find out the Reduction Product of 'A'

- (1) Iso pentyl amine
- (2) n-pentyl amine
- (3) Iso butyl amine
- (4) Tertiary pentyl amine

10. The amine produced by the Hoffmann degradation of benzamide is—



11. Boiling point order of isomeric amine is—

- (1) Primary > Secondary > Tertiary
 (2) Primary < Secondary < Tertiary
 (3) Primary > Secondary < Tertiary
 (4) Primary < Secondary > Tertiary

12. Which of the following have maximum value of pK_b in Aqueous phase—

- (1) N-Methyl aniline
 (2) N, N-diethyl ethanamine
 (3) Benzenamine
 (4) Phenyl methanamine

13. Correct order of basic nature is—

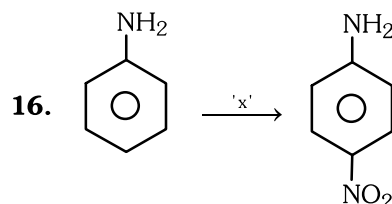
- (1) $(C_2H_5)_2NH > C_2H_5NH_2 > (C_2H_5)_3N > NH_3$
 (2) $(C_2H_5)_2NH > (C_2H_5)_3N > C_2H_5NH_2 > NH_3$
 (3) $(C_2H_5)_2NH < (C_2H_5)_3N < C_2H_5NH_2 < NH_3$
 (4) $NH_3 < (C_2H_5)_2NH < C_2H_5NH_2 < (C_2H_5)_3N$

14. Isocyanide test is used for—

- (1) Only Primary amine
 (2) Only secondary amine
 (3) Only chloroform
 (4) Both chloroform & Primary amine

15. Which one is soluble in NaOH :—

- (1) N-Ethyl benzenesulphonamide
 (2) N,N-Diethyl benzenesulphonamide
 (3) N-Ethyl aniline
 (4) N-Methyl ethanamine



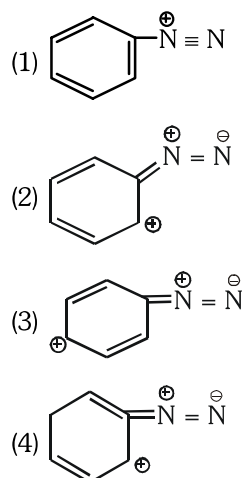
'x' in above reaction may be—

- (1) $H_2SO_4 / 288\text{ K}$
 (2) $CH_3COCl / HNO_3, H_2SO_4 / H_3O^+$
 (3) $HNO_3, H_2SO_4 / CH_3COCl, AlCl_3$
 (4) $H_3O^+ / HNO_3, H_2SO_4 / CH_3COCl$

17. Which one can exist as a zwitter ion—

- (1) Sulphamic acid
 (2) Sulphanilic Acid
 (3) Anilinium hydrogen sulphate
 (4) Aniline

18. Which of the following resonating structure is not correct—



19. Gattermann reaction is–

- (1) $\text{Ar N}_2 \text{ Br} + \text{CuCl/HCl} \longrightarrow \text{ArCl} + \text{N}_2$
(2) $\text{Ar N}_2 \text{ Br} + \text{CuCN/KCN} \longrightarrow \text{ArCN} + \text{N}_2$
(3) $\text{Ar N}_2 \text{ Br} + \text{Cu/HBr} \longrightarrow \text{Ar Br} + \text{N}_2$
(4) $\text{Ar N}_2 \text{ Br} + \text{KI} \longrightarrow \text{Ar I} + \text{N}_2$

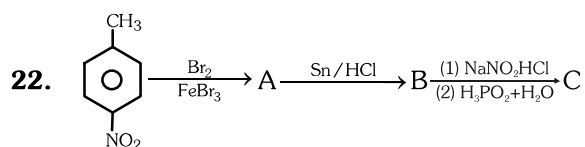
20. Benzene diazonium chloride can be converted into benzene by using–

- (1) $\text{H}_3\text{PO}_2 + \text{H}_2\text{O}$ (2) $\text{SnCl}_2 + \text{HCl}$
(3) H_3PO_4 (4) H_2O (warm)

21. Benzendiazonium chloride + 'x' $\xrightarrow{\text{OH}^-}$ Yellow dye

'x' in above reaction is

- (1) Phenol (2) Aniline
(3) Benzene (4) Benzene sulphonic acid



Product 'C' is–

